

MADHVIKA SEHGAL

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PROFESSIONAL SUMMARY

Business Intelligence Engineer with 4+ years of experience building scalable data pipelines, ETL workflows, and analytics solutions across healthcare and fintech domains. Skilled in SQL, Python, AWS, and Spark SQL, with experience supporting finance reporting, data quality, and operational analytics. Experienced leveraging Generative AI and LLM-assisted tooling to automate workflows and improve data governance.

TECHNICAL SKILLS

Languages & Query: SQL, Python (Pandas, NumPy), Spark SQL, SAS

Data & Visualization Platforms: AWS (Athena, Redshift, S3, Glue, Lambda), Oracle, Tableau Prep, Alteryx, Tableau, QuickSight

Data Engineering: Data Modeling, Data Warehousing, Data Quality & Reconciliation, Root Cause Analysis, Data Lineage, Pipeline Debugging, KPI Standardization, Workflow Automation

Pipeline & Orchestration: ETL/ELT Engineering, DAG-dependent workflow management, Apache Airflow (internal equivalent), dbt, CICD-managed pipeline versioning, Git

AI & Emerging: Generative AI (Amazon Bedrock), Agentic workflow automation

PROFESSIONAL EXPERIENCE

Business Intelligence Engineer | **Amazon - Pharmacy Finance & Analytics** | Seattle, WA | Aug 2025 - Present

- **Finance Allocation Pipeline Ownership:** Enhanced and validated a 1,900+ line Spark SQL cost allocation pipeline powering **FP&A-driven** Pharmacy Contribution Profit reporting and maintaining GL tie-back integrity across DAG-dependent pipelines.
- Re-designed **cost allocation logic for cost center restructuring**, analyzing downstream dependencies and validating allocation changes through **SQL reconciliation frameworks** to ensure accurate financial attribution across ops domains.
- Automated a manual workforce cost allocation process by integrating **ADP payroll, warehouse metadata, and business logic** into a SQL-based pipeline, eliminating manual S3 dependencies and reducing recurring month-end operational effort by 7+ hrs. monthly.
- **Governance Analytics & Platform Visibility:** Solved limited visibility into data platform adoption by reverse-engineering audit logs (**1M+ monthly events**) and building classification logic across datasets, users, and tools; delivered a governance dashboard that enabled adoption tracking and data-driven deprecation and capacity planning.
- **Built Kiosk P&L Framework** as a net-new cost entity across finance pipelines, built attribution flags, designed exclusion/inclusion logic across 5 cost pools, and surfaced kiosk costs end-to-end through the Contribution Profit pipeline, enabling Finance to independently evaluate kiosk unit economics for the first time.
- Managed **primary key migration across 10+ pipeline tables**, auditing join dependencies, resolving data quality blockers, and validating zero metric loss.

Business Intelligence Engineer | **Barclays - Credit Card, Contact Center Analytics** | Whippany, NJ | Jan 2021 - Jul 2025

- **Callback Feature Revamp:** Led a 3-member team to redesign SLA logic and integrate callback features into Contact Center workflows, driving process improvement across 45+ dashboards and reports, boosting queue efficiency & improving customer retention by 15%.
- **GenAI-Driven Automation:** Engineered a serverless solution using **AWS Lambda and Claude LLM via Bedrock** to automate report metadata summarization, significantly improving audit readiness and version control from days to minutes.
- **Cloud Migration & Data Lake Setup:** Co-led the migration of multi-terabyte datasets from **Oracle to AWS S3**, creating a scalable data lake that cut query latency by 40% using **Athena and Redshift**.
- **Customer Journey Analytics:** Analyzed clickstream, call log, and servicing data across 5M+ IVR interactions. Created heatmaps and predictive **Tableau** models to forecast demand and support capacity planning decisions, reducing call time by 15% and transfers by 20%.
- **ETL Framework for IVR Dashboards:** Built and managed end-to-end **ETL processes** powering over 30 IVR **Tableau dashboards**, enabling standardized data pipelines for real-time funnel monitoring and KPI consistency.
- **A/B Testing for Queue Optimization:** Led design and execution of **A/B tests for queue restructuring** (Simple vs Complex), boosting resolution efficiency by 12%, and developed a Tableau dashboard to track month-over-month performance.
- **Agent Performance Scoring Model:** Developed a scoring model using sentiment, CSAT, and NPS data, incorporating **Z-score normalization** to benchmark agent performance and identify outliers for QA and coaching.
- **Real-Time Marketing Insights:** Developed Redshift-powered dashboards supporting real-time **KPI tracking**, leading to a 20% increase in IVR containment and reducing manual reporting effort by 80%.
- **IVR Funnel Monitoring:** Built comprehensive dashboards for funnel KPIs like containment, self-service, and sentiment; introduced tagging improvements that cut call transfers by 30%.
- **Cross-Functional Enablement:** Delivered scalable analytics pipelines for product, Customer Experience (CX), Fraud-Disputes, and Collections teams. Used **Agile (JIRA/Confluence)** to manage workflows, increasing delivery speed by 30%.

Business Consultant | Accenture – SAP MM Consultant | Delhi, India | Jan 2017 – May 2019

- **Led full-cycle SAP MM implementations** for global clients, translating complex business requirements into technical specifications, managing stakeholder workshops, UAT, and post-go-live support with 95% user adoption.
- Facilitated cross-functional collaboration between business and IT teams to ensure **seamless integration** of MM with SD, FI, and PP modules, enhancing data integrity, inter-system communication, and **enterprise resource planning** continuity.
- Developed functional specifications and **business requirement documents (BRDs)** for custom reports and enhancements; partnered with ABAP developers to deliver scalable, logic-driven solutions aligned with business **KPIs and change management**.

EDUCATION

- **Master of Science, Management Information Systems, Focus – Business Intelligence and Data Analytics**
University of Arizona, Tucson, AZ (Aug 2019 – Dec 2020)
- **Bachelor of Technology, Information Technology**
Guru Gobind Singh Indraprastha University, New Delhi, India (August 2012- May 2016)

CERTIFICATIONS

- **Gen AI Product Management Specialization**, Duke University (Coursera, 2025)
- **Generative AI: Prompt Engineering Basics**, IBM (Coursera, 2025)